

SESSION 2

Hands-On Activity: Building a Baseline ML Model

Dataset inspection • Baselines • Simple models • Metrics • Interpretation

Applied AI Design Workshop for Faculty

Fred Livingston (fjliving@ncsu.edu) 2026

Goal: complete one defensible first-pass ML workflow, not just run a model.

Activity outputs

- Problem statement
- Baseline model
- Trained simple model
- Initial metrics
- Interpretation and limitation statement

Session 2 learning outcomes

Inspect

Check rows,
variables, types,
missing values, and
target definition

Frame

Decide whether the
dataset supports
regression or
classification

Baseline

Establish a simple
comparison model
before training
anything complex

Train

Fit one or two
simple models
using a transparent
workflow

Interpret

Compare metrics,
identify limitations,
and recommend a
next step

Participant output: a concise model card for a first-pass baseline ML analysis.

Activity roadmap



The purpose is to practice a complete, defensible workflow, **not to optimize model performance.**

Process Run Quality Dataset

A compact engineering dataset to practice the full workflow without domain-specific software barriers.

Run	Temp	Speed	Vibration	Material	Yield	Pass
1	185	42	0.18	A	88	Yes
2	190	46	0.35	B	72	No
3	174	39	0.22	A	91	Yes
4	205	52	0.48	C	61	No
5	198	44	0.20	B	86	Yes

Regression option
Predict continuous yield.

Classification option
Predict pass/fail quality.

Baseline required
Compare against simplest
useful rule.

Part 1: define the problem

Start by writing down what the model is expected to predict and how success will be judged.

Decision or prediction

What question should the model answer?

Example: Will this run pass quality inspection?

Unit of observation

What is one row?

Example: one process run, part, student, machine cycle, sample, or day.

Metric

What score reflects success?

Example: recall if missed defects are costly; MAE if error magnitude matters.

Target

What is the outcome?

Regression: number,
Classification: class label

one sentence problem statement + selected metric + baseline choice.

Part 2: inspect the data before modeling

A baseline model is only meaningful if the dataset has been checked for basic validity.

Rows and columns

Do we have enough observations for the number of variables?

Data types

Which features are numerical, categorical, date/time, or text?

Missing values

Are values missing randomly, systematically, or by design?

Target definition

Is the target measurable, consistent, and available after the prediction point?

Leakage risk

Would any input reveal the answer too directly or too late?

Split strategy

Should the data be split randomly, by group, or by time?

Mini-lesson: data leakage can invalidate results

Leakage occurs when the model uses information that would not be available at the time of prediction.

Example leakage feature

Final inspection score included as an input when predicting pass/fail quality.

The model is effectively seeing the answer.

Time leakage

Using future sensor readings when predicting an earlier failure or yield.

The timeline is broken.

Group leakage

Samples from the same subject, machine, or batch appear in both train and test sets.

The test set is too familiar.

Activity: Identify one variable that could cause leakage and justify keeping or removing it.

Part 3: establish the baseline

The baseline answers: what performance can we get with a simple, transparent rule?

Regression baseline

Predict the training-set mean or median.

Metrics: MAE, RMSE, R^2 relative to simple prediction.

Classification baseline

Predict the majority class or current rule.

Metrics: confusion matrix, accuracy, precision, recall, F1.

Why it matters

A model that does not beat the baseline should not be treated as useful, even if it is complex.

Part 4: train simple models

Use simple, interpretable models first. The goal is to understand the problem before chasing accuracy.

Regression path

1. Mean baseline
2. Linear regression
3. Decision-tree regression

Compare training and test error.

Classification path

1. Majority baseline
2. Logistic regression
3. Decision tree

Compare confusion matrices.

Model discipline

Use the same split, same inputs, and same target for each comparison.

Change one thing at a time.

Part 5: evaluate results

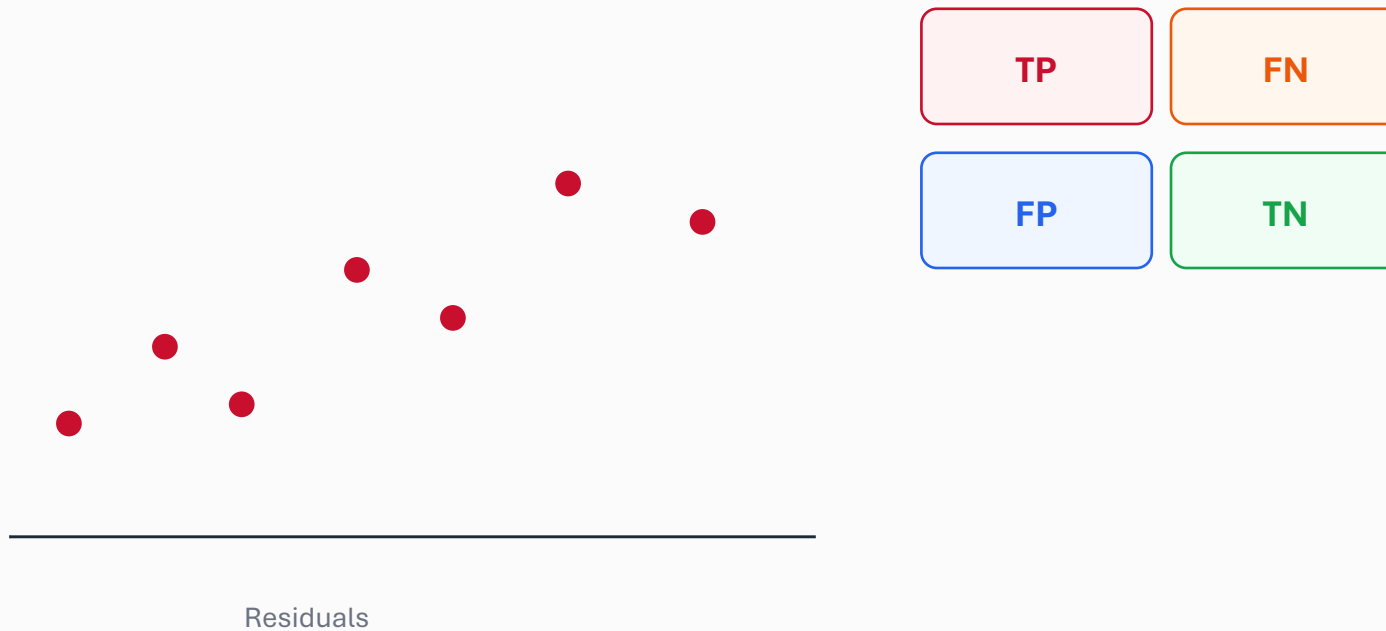
Evaluation should answer both “How well did it predict?” and “What kinds of errors did it make?”

Regression	MAE	Average absolute error in original units
Regression	RMSE	Penalizes larger errors more strongly
Regression	R^2	Fraction of target variation explained
Classification	Accuracy	Overall percent correct
Classification	Recall	Fraction of important positives detected
Classification	Precision	Fraction of predicted positives that were correct

Metric choice should reflect the practical consequence of the error, not just mathematical convenience.

Part 5 continued: inspect error patterns

A first model is most valuable when it shows where the data or problem formulation needs improvement.



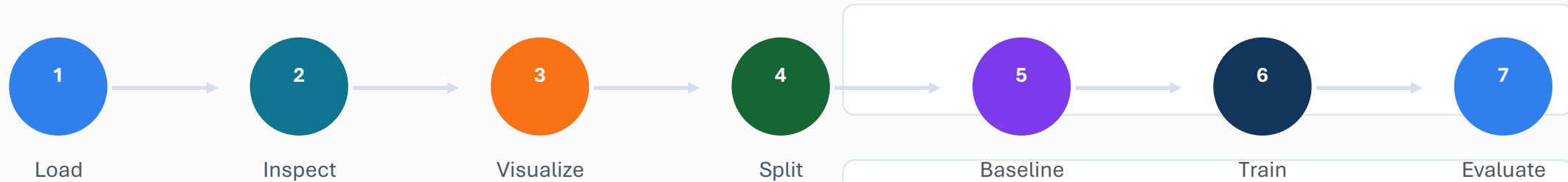
Questions

- Are errors larger for one group?
- Are mistakes tied to missing features?
- Is one class poorly represented?
- Is the target noisy?

Error analysis converts model performance into research and data-collection priorities.

Session 2 technical skill: complete baseline modeling workflow

Goal: combine the tool skills into one repeatable applied ML workflow.



```
df = pd.read_csv("classification_workshop.csv")
X = df[features]
y = df[target]

X_train, X_test, y_train, y_test = train_test_split(X, y, stratify=y)

model.fit(X_train, y_train)
pred = model.predict(X_test)
```

Questions?

- Are errors larger for one group?
- Are mistakes tied to missing features?
- Is one class poorly represented?
- Is the target noisy?

What faculty produce

- A baseline comparison
- A first model result
- A metric-based interpretation
- One limitation
- One recommended next step

Session 2 notebook

02_session2_complete_workflow.ipynb

Participant output: first-pass model card

Each group should leave with a short model card summarizing the baseline analysis.

Problem statement

What is predicted, for whom or what, and why?

Dataset

Rows, variables, target, missing values, and leakage concerns

Baseline

Simple rule and its performance

Model

Algorithm, inputs, split method, and metric

Results

Performance compared with baseline and observed error pattern

Limitations

What should not be concluded and what data are needed next?

Discussion prompts

Did the model outperform the baseline?

Was the improvement practically meaningful?

Which metric best represents success?

Did the model appear to overfit?

Were any variables likely to introduce leakage?

Would the model generalize to another lab, machine, course, or population?

What additional data should be collected?

What limitation must be reported?

Synthesis: a baseline workflow is a scientific argument about whether the data support a useful prediction.

Session 2 wrap-up: what you should take forward

A complete first-pass ML workflow should be transparent, reproducible in class, and easy to critique.

Before modeling

- Define target
- Define unit
- Select metric
- Inspect data
- Check leakage

During modeling

- Establish baseline
- Use simple models
- Keep comparison fair
- Evaluate on held-out data

After modeling

- Interpret errors
- State limitations
- Recommend next data
- Avoid causal overclaims

Day 4 endpoint: You can frame an applied ML problem and run a defensible baseline analysis.